

DR. D. Y. PATIL INSTITUTE OF ENGINEERING, MANAGEMENT AND RESEARCH
Akurdi, Pune-411044

Department-Artificial Intelligence and Data Science

Laboratory Journal | **Subject:** Computer Laboratory – IV

AY:2025-26

Name: Atharva Velhankar

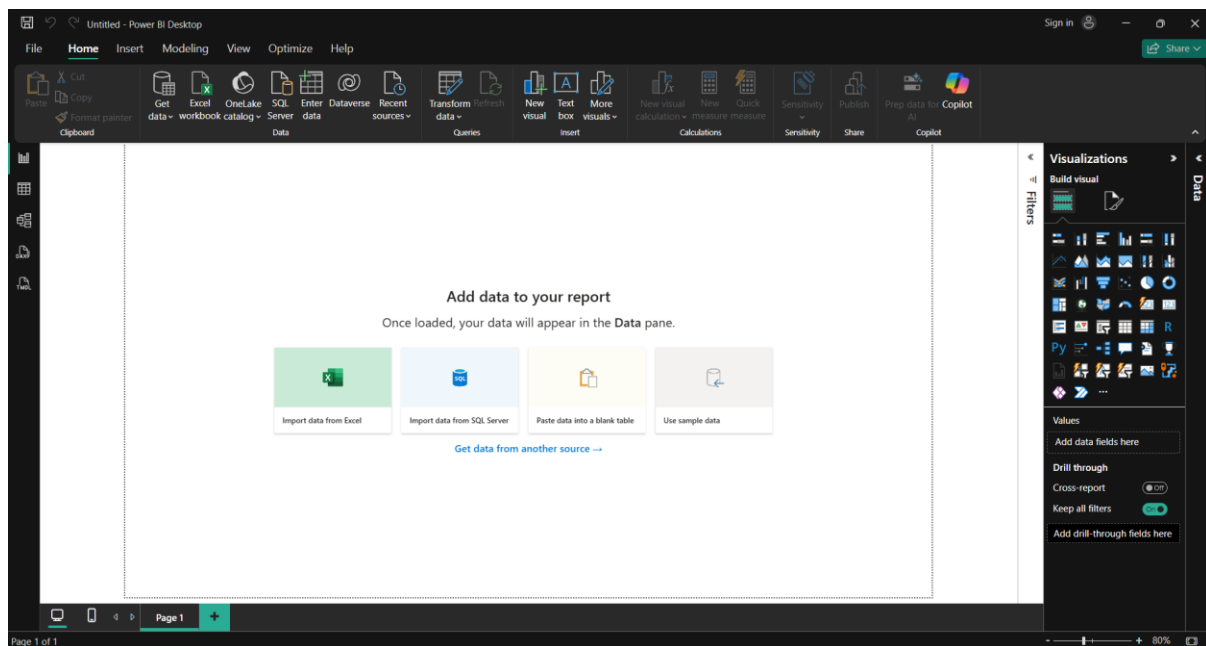
Class: B.E. (A)

Roll No: 22508

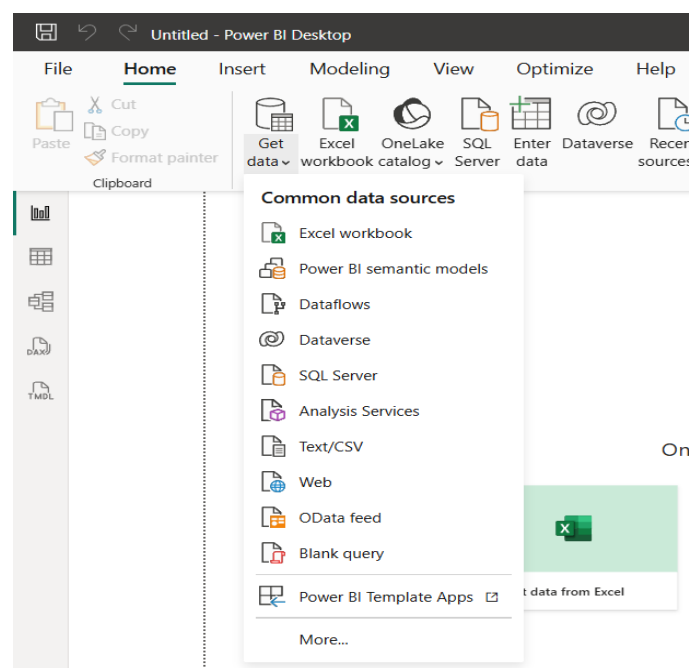
Lab Practical

Import Data from different Sources such as (Excel, Sql Server, Oracle etc.) and load in targeted system.

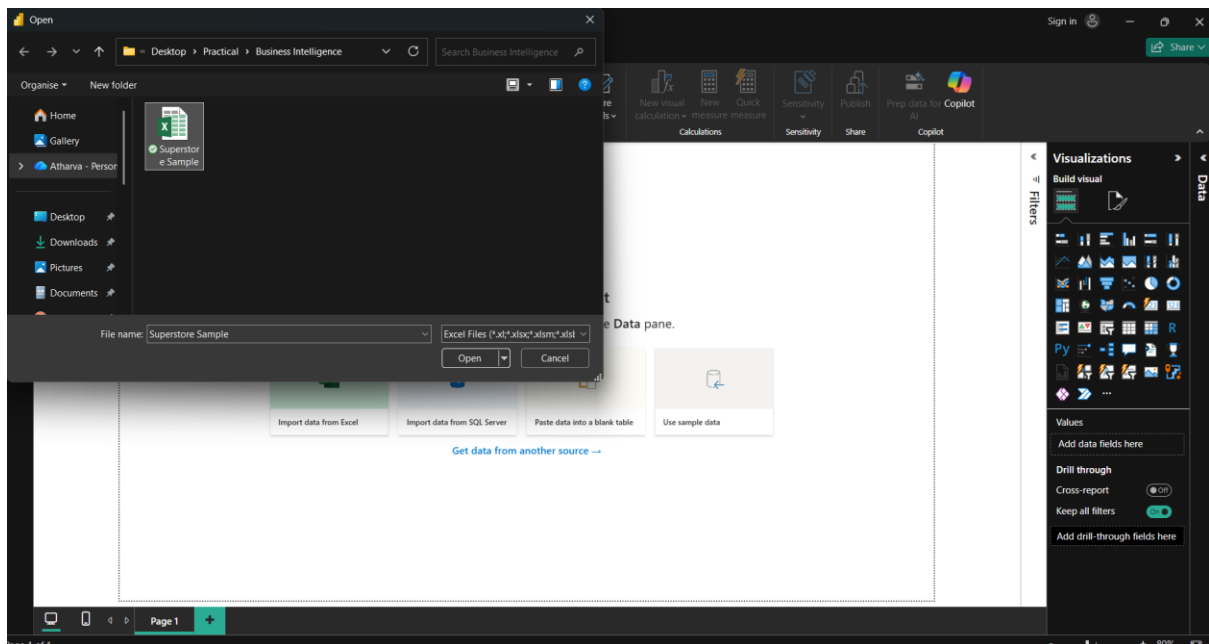
1) Launch Power BI Desktop.



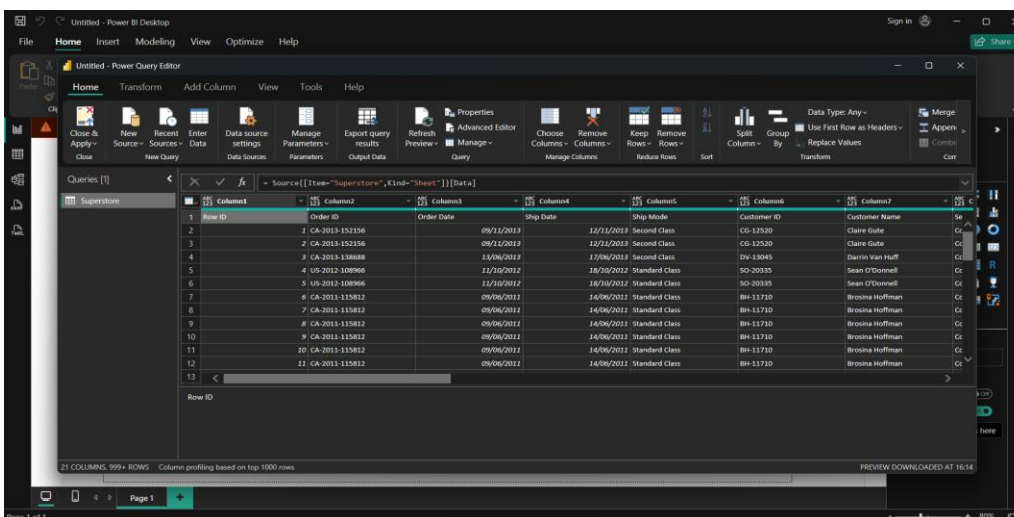
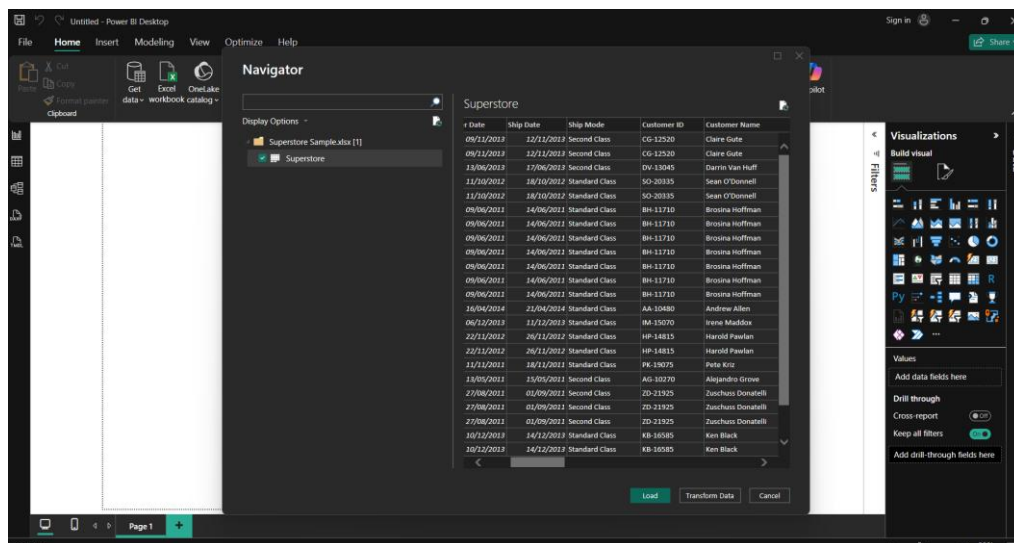
2) From the Home ribbon, select Get Data.



3) In the Open File dialog box, select the Products.xlsx file.



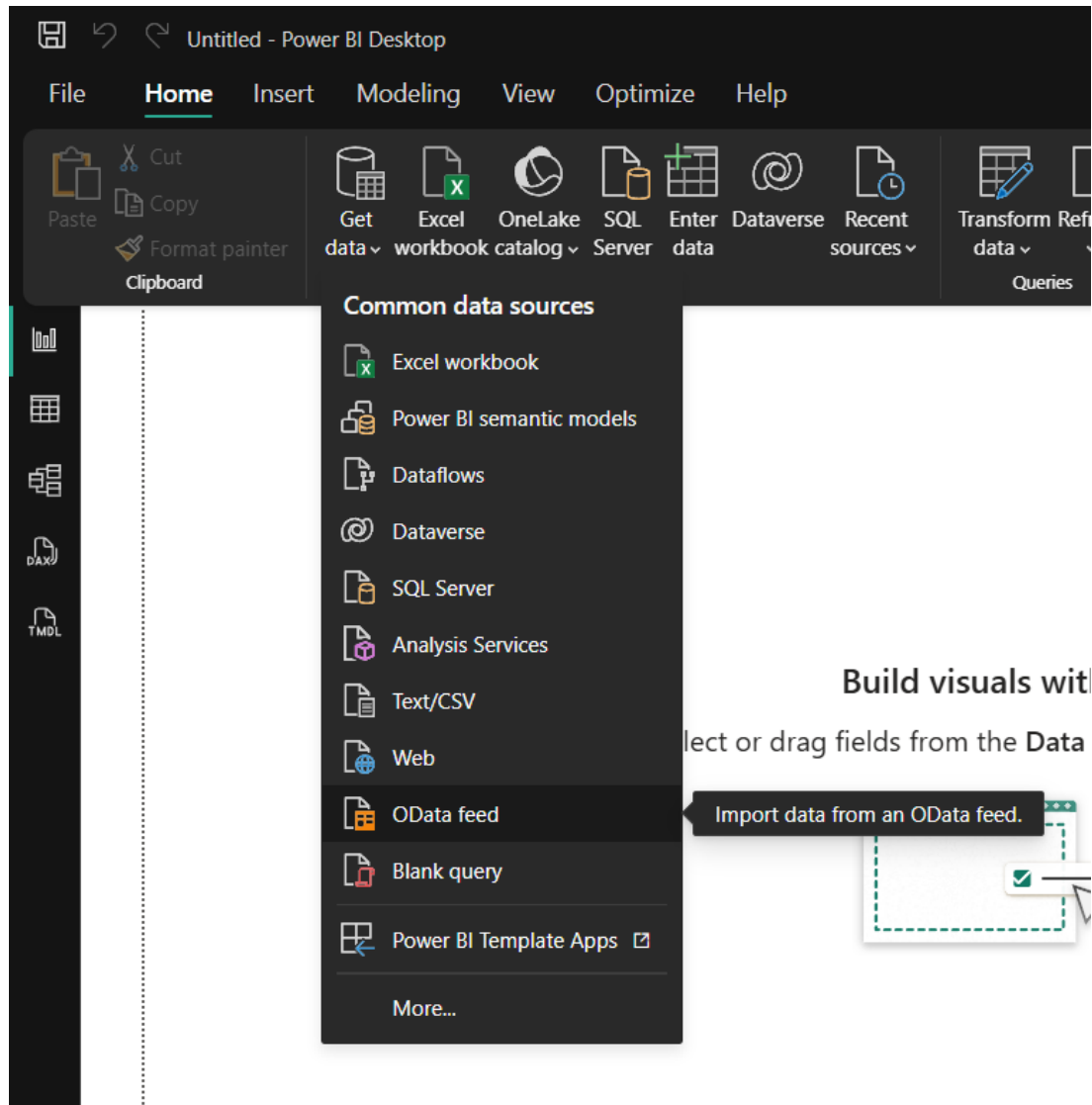
4) In the Navigator pane, select the Products table and then select Edit.



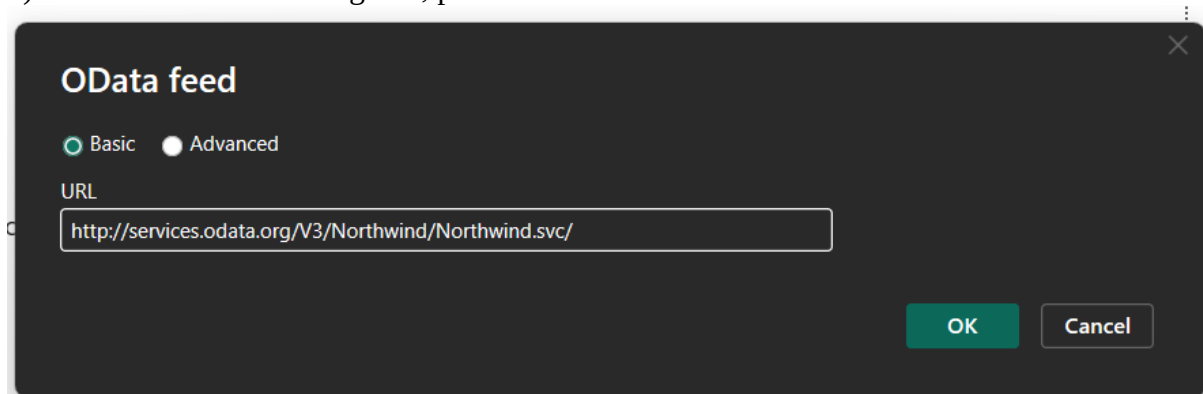
Importing Data from OData Feed

(<http://services.odata.org/V3/Northwind/Northwind.svc/>)

1) From the Home ribbon tab in Query Editor, select Get Data. Browse to the OData Feed data source.



2) In the OData Feed dialog box, paste the URL for the Northwind OData feed.



3) In the Navigator pane, select the Orders table, and then select Edit

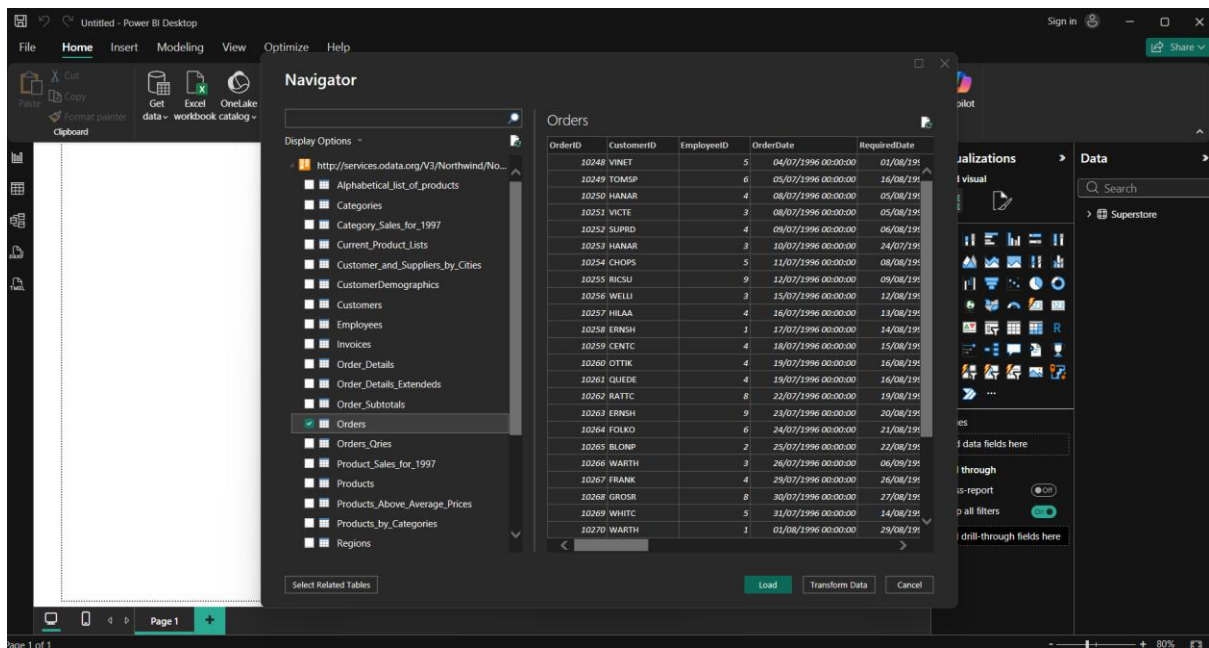


Table: Orders (830 rows)

OrderID	CustomerID	EmployeeID	OrderDate	RequiredDate	ShippedDate	ShipVia	Freight	ShipName	ShipAddress	ShipCity	ShipRegion	ShipPost
10249	TOMSP	6	05/07/1996 00:00:00	16/08/1996 00:00:00	10/07/1996 00:00:00	1	11.61	Toms Spezialitäten	Luisestr. 48	Münster		44807
10260	OTTIK	4	19/07/1996 00:00:00	16/08/1996 00:00:00	29/07/1996 00:00:00	1	55.09	Ottiles Käseladen	Mehrheimerstr. 369	Köln		50739
10267	FRANK	4	29/07/1996 00:00:00	26/08/1996 00:00:00	06/08/1996 00:00:00	1	208.58	Frankenversand	Berliner Platz 43	München		80805
10273	QUICK	3	05/08/1996 00:00:00	02/09/1996 00:00:00	12/08/1996 00:00:00	3	76.07	QUICK-Stop	Taucherstraße 10	Cunewalde		01307
10277	MORIG	2	09/08/1996 00:00:00	06/09/1996 00:00:00	13/08/1996 00:00:00	3	125.77	Morgenstern Gesundkost	Heerstr. 22	Leipzig		04179
10279	LEHMS	8	13/08/1996 00:00:00	10/09/1996 00:00:00	16/08/1996 00:00:00	2	25.83	Lehmanns Marktstand	Magazinweg 7	Frankfurt a.M.		60528
10284	LEHMS	4	19/08/1996 00:00:00	16/09/1996 00:00:00	27/08/1996 00:00:00	1	76.56	Lehmanns Marktstand	Magazinweg 7	Frankfurt a.M.		60528
10285	QUICK	7	20/08/1996 00:00:00	17/09/1996 00:00:00	26/08/1996 00:00:00	2	76.83	QUICK-Stop	Taucherstraße 10	Cunewalde		01307
10286	QUICK	8	21/08/1996 00:00:00	18/09/1996 00:00:00	30/08/1996 00:00:00	3	229.24	QUICK-Stop	Taucherstraße 10	Cunewalde		01307
10301	WANDK	8	09/09/1996 00:00:00	07/10/1996 00:00:00	17/09/1996 00:00:00	2	45.08	Die Wandernde Kuh	Adenauerallee 900	Stuttgart		70563
10312	WANDK	2	23/09/1996 00:00:00	21/10/1996 00:00:00	03/10/1996 00:00:00	2	40.26	Die Wandernde Kuh	Adenauerallee 900	Stuttgart		70563
10313	QUICK	2	24/09/1996 00:00:00	22/10/1996 00:00:00	04/10/1996 00:00:00	2	1.96	QUICK-Stop	Taucherstraße 10	Cunewalde		01307
10323	KOENE	4	07/10/1996 00:00:00	04/11/1996 00:00:00	14/10/1996 00:00:00	1	4.88	Königlich Essen	Maubelstr. 90	Brandenburg		14776
10325	KOENE	7	09/10/1996 00:00:00	23/10/1996 00:00:00	14/10/1996 00:00:00	3	64.86	Königlich Essen	Maubelstr. 90	Brandenburg		14776
10337	FRANK	4	24/10/1996 00:00:00	21/11/1996 00:00:00	29/10/1996 00:00:00	3	108.26	Frankenversand	Berliner Platz 43	München		80805
10342	FRANK	4	30/10/1996 00:00:00	13/11/1996 00:00:00	04/11/1996 00:00:00	2	54.83	Frankenversand	Berliner Platz 43	München		80805
10343	LEHMS	4	31/10/1996 00:00:00	28/11/1996 00:00:00	06/11/1996 00:00:00	1	110.37	Lehmanns Marktstand	Magazinweg 7	Frankfurt a.M.		60528
10345	QUICK	2	04/11/1996 00:00:00	02/12/1996 00:00:00	11/11/1996 00:00:00	2	249.06	QUICK-Stop	Taucherstraße 10	Cunewalde		01307
10348	WANDK	4	07/11/1996 00:00:00	05/12/1996 00:00:00	15/11/1996 00:00:00	2	0.78	Die Wandernde Kuh	Adenauerallee 900	Stuttgart		70563
10356	WANDK	6	18/11/1996 00:00:00	16/12/1996 00:00:00	27/11/1996 00:00:00	2	36.71	Die Wandernde Kuh	Adenauerallee 900	Stuttgart		70563
10361	QUICK	7	22/11/1996 00:00:00	20/12/1996 00:00:00	03/12/1996 00:00:00	2	183.17	QUICK-Stop	Taucherstraße 10	Cunewalde		01307
10363	DRACD	4	26/11/1996 00:00:00	24/12/1996 00:00:00	04/12/1996 00:00:00	3	30.54	Drachenblut Delikatessen	Walsenweg 21	Aachen		52066
10391	DRACD	3	23/12/1996 00:00:00	20/01/1997 00:00:00	31/12/1996 00:00:00	3	5.45	Drachenblut Delikatessen	Walsenweg 21	Aachen		52066
10396	FRANK	1	27/12/1996 00:00:00	10/01/1997 00:00:00	06/01/1997 00:00:00	3	135.35	Frankenversand	Berliner Platz 43	München		80805
10407	OTTIK	2	07/01/1997 00:00:00	04/02/1997 00:00:00	30/01/1997 00:00:00	2	91.48	Ottiles Käseladen	Mehrheimerstr. 369	Köln		50739
10418	QUICK	4	17/01/1997 00:00:00	14/02/1997 00:00:00	24/01/1997 00:00:00	1	17.55	QUICK-Stop	Taucherstraße 10	Cunewalde		01307

Sign: _____.

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Class: B.E. (A)

Roll No: 22508

Lab Practical

Data Visualization from Extraction Transformation and Loading (ETL) Process.

Step 1: Check Data (Extraction)

```
import pandas as pd
import matplotlib.pyplot as plt

# EXTRACT - Load raw data from CSV
df = pd.read_csv("Superstore.csv", encoding='latin1')

print("Shape of data:", df.shape)
print("\nFirst 5 rows:")
df.head()
```

✓ 5.2s

Shape of data: (9994, 21)

Step 2: Transform

Check-missing values

```
# Check missing values
print("Missing values:\n", df.isnull().sum())

# Drop rows with null values
df = df.dropna()

# Select only required columns
df = df[['Category', 'Sub-Category', 'Sales', 'Profit', 'Quantity']]

✓ 0.0s
```

Missing values:

Row ID	0
Order ID	0
Order Date	0
Ship Date	0
Ship Mode	0
Customer ID	0
Customer Name	0
Segment	0
Country	0
City	0
State	0
Postal Code	0
Region	0
Product ID	0
Category	0
Sub-Category	0
Product Name	0
Sales	0
Quantity	0
Discount	0
Profit	0

dtype: int64

Remove Duplicates if any

```
df = df.drop_duplicates()

print("\nCleaned Data Shape:", df.shape)
print(df.head())
```

✓ 0.3s

Cleaned Data Shape: (7729, 5)

	Category	Sub-Category	Sales	Profit	Quantity
0	Furniture	Bookcases	261.9600	41.9136	2
1	Furniture	Chairs	731.9400	219.5820	3
2	Office Supplies	Labels	14.6200	6.8714	2
3	Furniture	Tables	957.5775	-383.0310	5
4	Office Supplies	Storage	22.3680	2.5164	2

Step 3: Load

```
# LOAD - Save cleaned data to new CSV
df.to_csv("Superstore_cleaned.csv", index=False)

print("Cleaned data loaded/saved successfully!")
```

✓ 0.0s

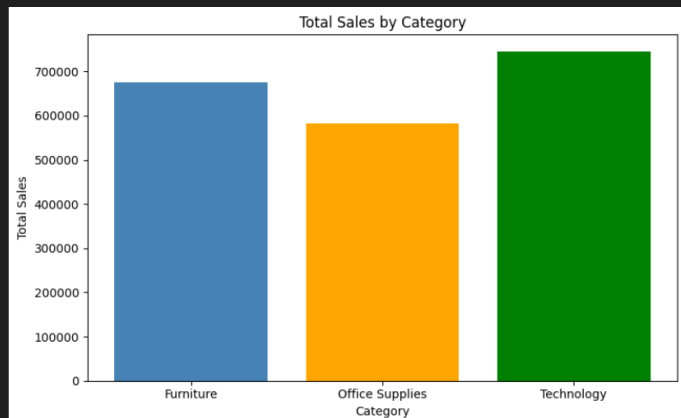
Cleaned data loaded/saved successfully!

Step 4: Visualization

```
category_sales = df.groupby('Category')['Sales'].sum()

plt.figure(figsize=(8,5))
plt.bar(category_sales.index, category_sales.values, color=['steelblue', 'orange', 'green'])
plt.title('Total Sales by Category')
plt.xlabel('Category')
plt.ylabel('Total Sales')
plt.tight_layout()
plt.show()
```

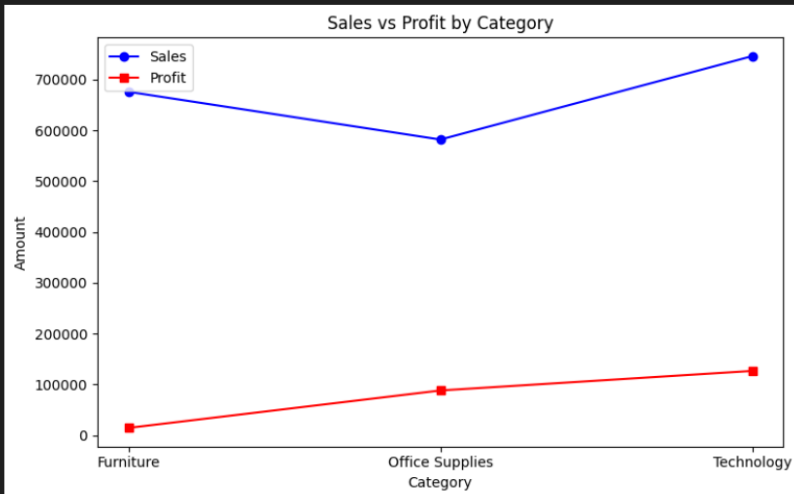
✓ 0.3s



```
category_profit = df.groupby('Category')['Profit'].sum()

plt.figure(figsize=(8,5))
plt.plot(category_sales.index, category_sales.values, marker='o', label='Sales', color='blue')
plt.plot(category_profit.index, category_profit.values, marker='s', label='Profit', color='red')
plt.title('Sales vs Profit by Category')
plt.xlabel('Category')
plt.ylabel('Amount')
plt.legend()
plt.tight_layout()
plt.show()
```

✓ 1.0s



Sign: _____.

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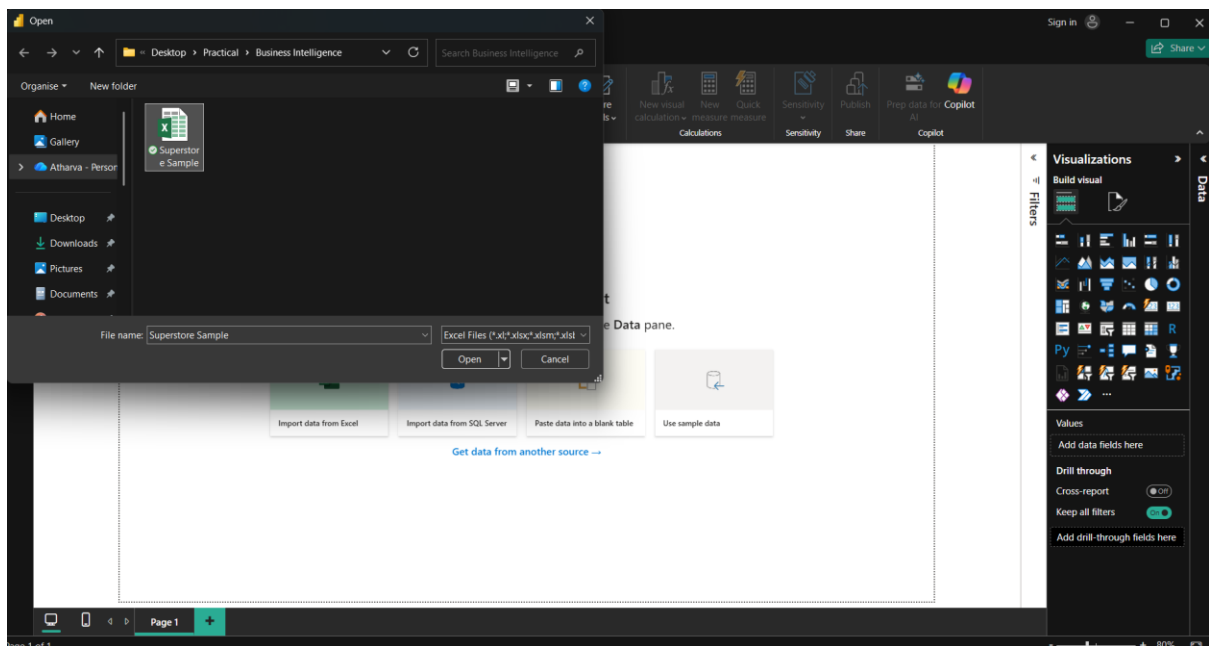
Class: B.E. (A)

Roll No: 22508

Lab Practical

Perform the Extraction Transformation and Loading (ETL) process to construct the database in the Sql server / Power BI.

STEP 1: Extraction (Load Data)



Navigator

Display Options

- Superstore Sample.xlsx [1]
- Superstore

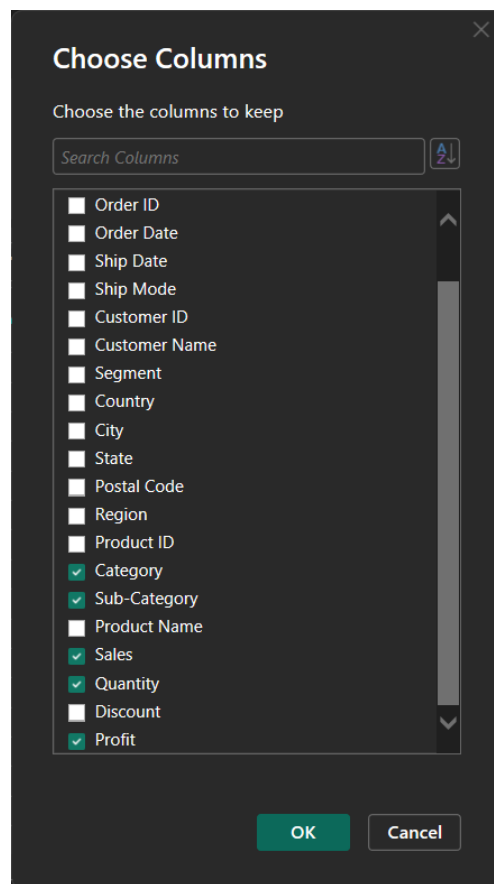
Superstore

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Cu
1	CA-2013-152156	09/11/2013	12/11/2013	Second Class	
2	CA-2013-152156	09/11/2013	12/11/2013	Second Class	
3	CA-2013-138688	13/06/2013	17/06/2013	Second Class	
4	US-2012-108966	11/10/2012	18/10/2012	Standard Class	
5	US-2012-108966	11/10/2012	18/10/2012	Standard Class	
6	CA-2011-115812	09/06/2011	14/06/2011	Standard Class	
7	CA-2011-115812	09/06/2011	14/06/2011	Standard Class	
8	CA-2011-115812	09/06/2011	14/06/2011	Standard Class	
9	CA-2011-115812	09/06/2011	14/06/2011	Standard Class	
10	CA-2011-115812	09/06/2011	14/06/2011	Standard Class	
11	CA-2011-115812	09/06/2011	14/06/2011	Standard Class	
12	CA-2011-115812	09/06/2011	14/06/2011	Standard Class	
13	CA-2014-114412	16/04/2014	21/04/2014	Standard Class	
14	CA-2013-161389	06/12/2013	11/12/2013	Standard Class	
15	US-2012-118983	22/11/2012	26/11/2012	Standard Class	
16	US-2012-118983	22/11/2012	26/11/2012	Standard Class	
17	CA-2011-105893	11/11/2011	18/11/2011	Standard Class	
18	CA-2011-167164	13/05/2011	15/05/2011	Second Class	
19	CA-2011-143336	27/08/2011	01/09/2011	Second Class	
20	CA-2011-143336	27/08/2011	01/09/2011	Second Class	
21	CA-2011-143336	27/08/2011	01/09/2011	Second Class	
22	CA-2013-137330	10/12/2013	14/12/2013	Standard Class	
23	CA-2013-137330	10/12/2013	14/12/2013	Standard Class	

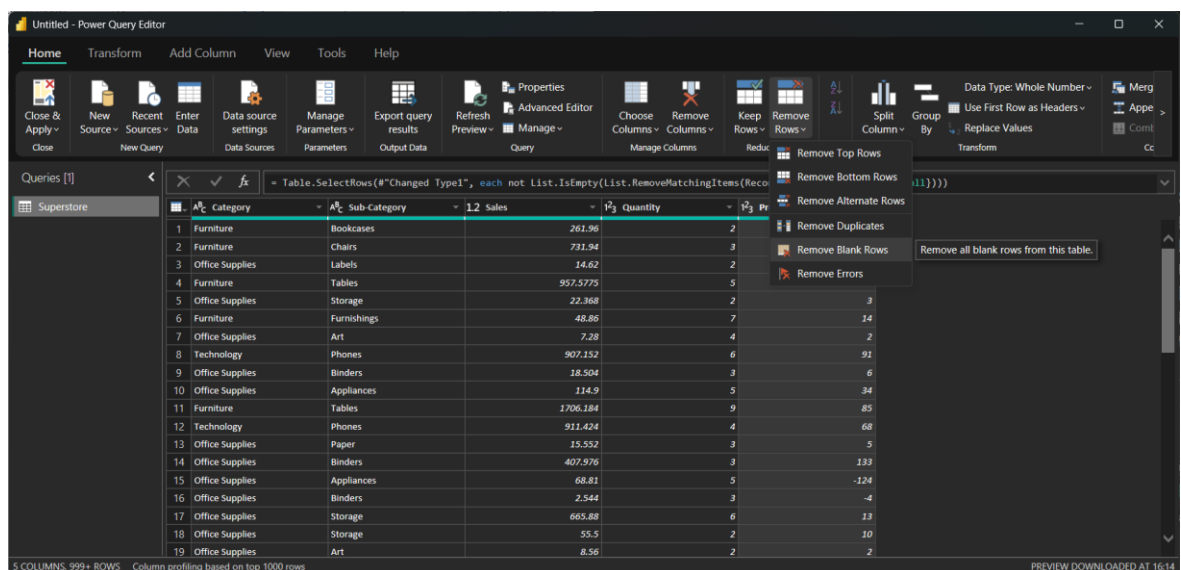
Load Transform Data Cancel

STEP 2: Transformation

(A) Remove Unnecessary Columns (Select the ones you want to keep)



(B) Handle Missing Values



(C) Changing the Data Type:

The screenshot shows the Power Query Editor interface. The 'Data Type' dropdown menu is open for the 'Sales' column, showing options like 'Decimal Number', 'Fixed decimal number', 'Whole Number', 'Percentage', 'Date/Time', 'Date', 'Time', 'Date/Time/Timezone', 'Duration', 'Text', 'True/False', and 'Binary'. The 'Sales' column is currently set to 'Decimal Number'.

Category	Sub-Category	Sales	Quantity	Profit
Furniture	Bookcases	261.96	2	41.9136
Furniture	Chairs	731.94	3	219.582
Office Supplies	Labels	14.62	2	6.8714
Furniture	Tables	957.5775	5	-383.031
Office Supplies	Storage	22.368	2	2.5164
Furniture	Furnishings	48.86	7	14.1694
Office Supplies	Art	7.28	4	1.9656
Technology	Phones	907.152	6	90.7152
Office Supplies	Binders	18.504	3	5.7825
Office Supplies	Appliances	114.9	5	34.47
Furniture	Tables	1706.184	9	85.3092
Technology	Phones	911.424	4	68.3568
Office Supplies	Paper	15.552	3	5.4432
Office Supplies	Binders	407.976	3	132.5922
Office Supplies	Appliances	68.81	5	-123.858
Office Supplies	Binders	2.544	3	-3.816
Office Supplies	Storage	665.88	6	13.3176
Office Supplies	Storage	55.5	2	9.99
Office Supplies	Art	8.56	2	2.4824

(D) Create NEW Column

The screenshot shows the Power Query Editor interface with the 'Add Column' tab selected. The 'Custom Column' dialog box is open, showing the formula `= [Profit] / [Sales] * 100` and the 'Available columns' list.

Category	Sub-Category	Sales	Quantity	Profit
Furniture	Bookcases	261.96	2	42
Furniture	Chairs	731.94	3	220
Office Supplies	Labels	14.62	2	7
Furniture	Tables	957.5775	5	-383
Office Supplies	Storage	22.368	2	3
Furniture	Furnishings	48.86	7	14
Office Supplies	Art	7.28	4	2
Technology	Phones	907.152	6	91
Office Supplies	Binders	18.504	3	6
Office Supplies	Appliances	114.9	5	34
Furniture	Tables	1706.184	9	85
Technology	Phones	911.424	4	68
Office Supplies	Paper	15.552	3	5
Office Supplies	Binders	407.976	3	133
Office Supplies	Appliances	68.81	5	-124
Office Supplies	Binders	2.544	3	-4
Office Supplies	Storage	665.88	6	13
Office Supplies	Storage	55.5	2	10
Office Supplies	Art	8.56	2	2

Custom Column

Add a column that is computed from the other columns.

New column name

Profit Margin

Custom column formula

`= [Profit] / [Sales] * 100`

Available columns

- Category
- Sub-Category
- Sales
- Quantity
- Profit

<< Insert

Learn about Power Query formulas

✓ No syntax errors have been detected.

OK Cancel

STEP 3: Loading

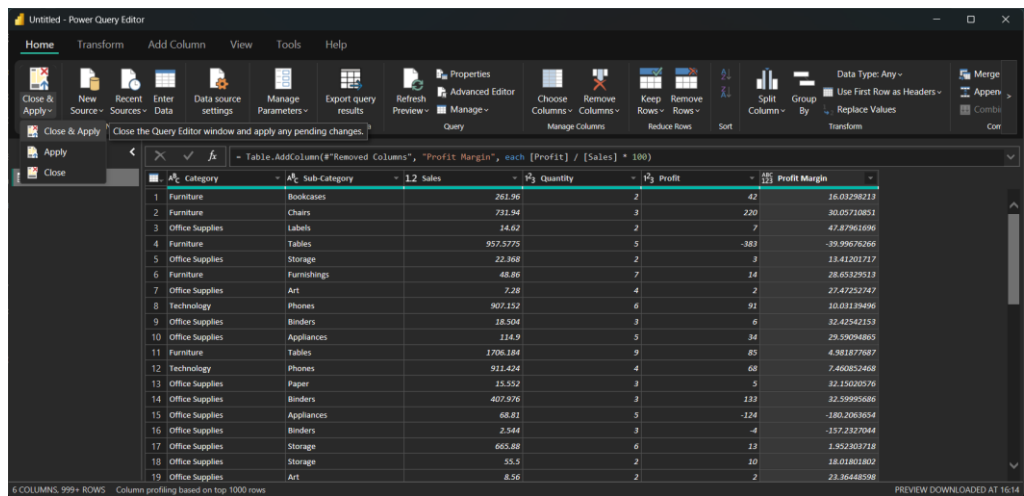


Table: AddColumn(8"Removed Column", "Profit Margin", each [Profit] / [Sales] * 100)

	Category	Sub-Category	Sales	Quantity	Profit	Profit Margin
1	Furniture	Bookcases	261.96	2	42	16.03298213
2	Furniture	Chairs	731.94	3	220	30.05710851
3	Office Supplies	Labels	14.62	2	7	47.87961696
4	Furniture	Tables	957.5775	5	-383	-39.99676266
5	Office Supplies	Storage	22.368	2	3	11.41201717
6	Furniture	Furnishings	48.86	7	14	28.65329513
7	Office Supplies	Art	7.28	4	2	27.47257747
8	Technology	Phones	807.152	6	91	10.03139496
9	Office Supplies	Binders	16.504	3	6	32.42421151
10	Office Supplies	Appliances	114.9	5	38	29.59094865
11	Furniture	Tables	1706.184	9	83	4.881877687
12	Technology	Phones	911.424	4	68	7.460252468
13	Office Supplies	Paper	15.552	3	5	32.15020576
14	Office Supplies	Binders	407.976	3	133	32.59995686
15	Office Supplies	Appliances	68.81	5	-124	-180.2063654
16	Office Supplies	Binders	2.544	3	-4	-157.2327044
17	Office Supplies	Storage	665.88	6	13	1.952303718
18	Office Supplies	Storage	55.3	2	10	18.01801802
19	Office Supplies	Art	8.56	2	2	23.36448598

6 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 16:14

STEP 4: Database Construction

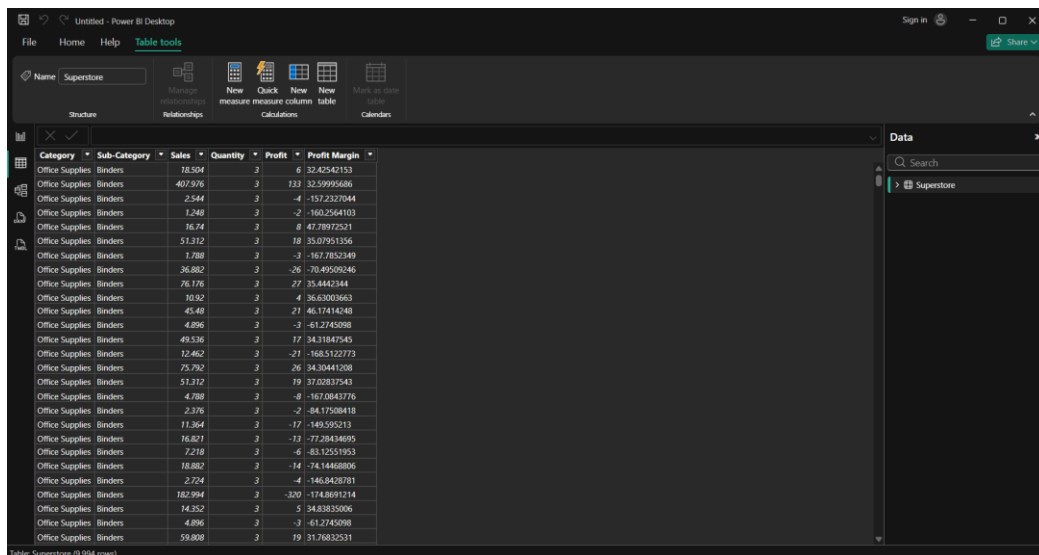


Table: Superstore (9,994 rows)

Category	Sub-Category	Sales	Quantity	Profit	Profit Margin
Office Supplies	Binders	18.504	3	6	32.42421151
Office Supplies	Binders	407.976	3	133	32.59995686
Office Supplies	Binders	2.544	3	-4	-157.2327044
Office Supplies	Binders	1.248	3	-2	-160.2564103
Office Supplies	Binders	16.74	3	8	47.78972521
Office Supplies	Binders	51.312	3	18	35.07951356
Office Supplies	Binders	1.788	3	-3	-167.7852349
Office Supplies	Binders	36.882	3	-26	-70.49509246
Office Supplies	Binders	26.176	3	27	35.4442144
Office Supplies	Binders	10.82	3	4	36.63023652
Office Supplies	Binders	45.48	3	21	46.17414248
Office Supplies	Binders	4.896	3	-3	-61.2745098
Office Supplies	Binders	49.536	3	17	34.31847545
Office Supplies	Binders	12.462	3	-21	-168.5122773
Office Supplies	Binders	75.792	3	26	34.30441208
Office Supplies	Binders	51.312	3	19	37.02837543
Office Supplies	Binders	4.788	3	-8	-162.0443776
Office Supplies	Binders	2.316	3	-2	-84.17508418
Office Supplies	Binders	11.364	3	-17	-149.595213
Office Supplies	Binders	16.821	3	-13	-77.28434695
Office Supplies	Binders	7.218	3	-6	-83.12551953
Office Supplies	Binders	18.882	3	-14	-74.14468806
Office Supplies	Binders	2.724	3	-4	-146.8426781
Office Supplies	Binders	182.994	3	-320	-174.8691214
Office Supplies	Binders	14.252	3	5	34.83838206
Office Supplies	Binders	4.896	3	-3	-61.2745098
Office Supplies	Binders	59.808	3	19	31.76832531

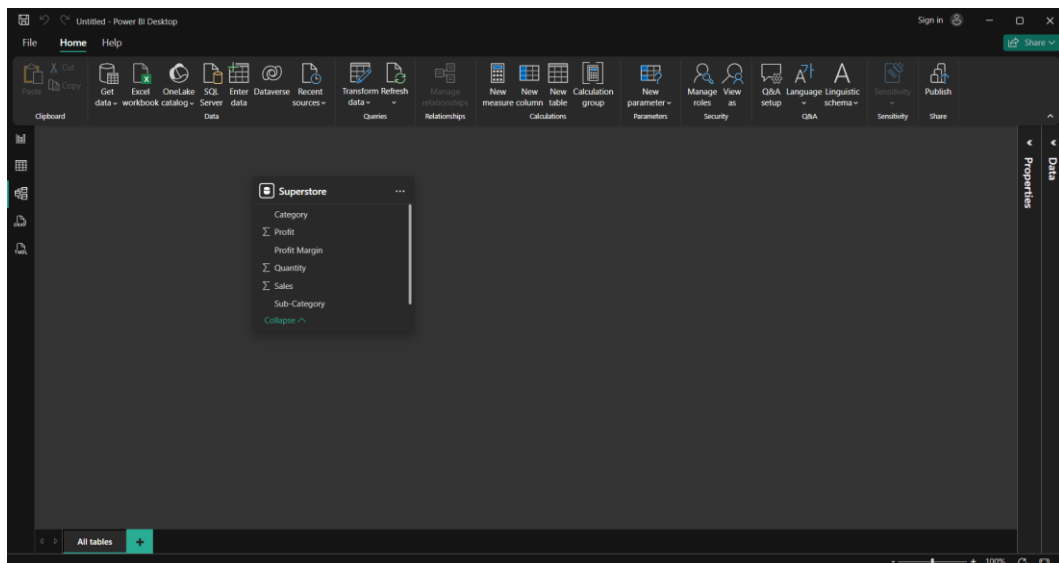


Table: Superstore (9,994 rows)

Category	Sub-Category	Sales	Quantity	Profit	Profit Margin
Office Supplies	Binders	18.504	3	6	32.42421151
Office Supplies	Binders	407.976	3	133	32.59995686
Office Supplies	Binders	2.544	3	-4	-157.2327044
Office Supplies	Binders	1.248	3	-2	-160.2564103
Office Supplies	Binders	16.74	3	8	47.78972521
Office Supplies	Binders	51.312	3	18	35.07951356
Office Supplies	Binders	1.788	3	-3	-167.7852349
Office Supplies	Binders	36.882	3	-26	-70.49509246
Office Supplies	Binders	26.176	3	27	35.4442144
Office Supplies	Binders	10.82	3	4	36.63023652
Office Supplies	Binders	45.48	3	21	46.17414248
Office Supplies	Binders	4.896	3	-3	-61.2745098
Office Supplies	Binders	49.536	3	17	34.31847545
Office Supplies	Binders	12.462	3	-21	-168.5122773
Office Supplies	Binders	75.792	3	26	34.30441208
Office Supplies	Binders	51.312	3	19	37.02837543
Office Supplies	Binders	4.788	3	-8	-162.0443776
Office Supplies	Binders	2.316	3	-2	-84.17508418
Office Supplies	Binders	11.364	3	-17	-149.595213
Office Supplies	Binders	16.821	3	-13	-77.28434695
Office Supplies	Binders	7.218	3	-6	-83.12551953
Office Supplies	Binders	18.882	3	-14	-74.14468806
Office Supplies	Binders	2.724	3	-4	-146.8426781
Office Supplies	Binders	182.994	3	-320	-174.8691214
Office Supplies	Binders	14.252	3	5	34.83838206
Office Supplies	Binders	4.896	3	-3	-61.2745098
Office Supplies	Binders	59.808	3	19	31.76832531

Sign: _____.